

Product Requirements Document: KOHLER AI Bathroom Designer & Planner

Track	Track 1, KOHLER-MITWPU AI Lab Case Study
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Repo	https://github.com/poojawork102/kohler-ai-bathroom-designer
Version	1.0 (19 Sept 2026)
Status legend	 Built ·  Partial ·  Planned

1. Summary

A web tool that turns a plain-English request ("8x10 ft bathroom, ₹6L budget, Japandi, smart features") into a **budget-compliant, space-valid, shoppable Kohler bathroom bundle** with a scaled 2D floorplan, itemised bill of materials, and sustainability metrics.

Design principle: the LLM only *understands* the request. All maths (cost, footprint, clearance) is done by deterministic code, so the output can never violate the budget or the room.

2. Problem

Pain point	Who feels it	Today
Fixtures don't fit the room (footprint, clearances, door swing)	Homeowners, designers	Found out at install
Style mismatch across toilet / shower / faucet / vanity	Homeowners	Needs a paid designer
Bundle exceeds budget	Homeowners	Manual spreadsheet
No visibility of water / energy impact	Eco-conscious buyers	Not shown at purchase

3. Goals and non-goals

Goals

- 1. Prompt → valid bundle + floorplan in under 2 seconds.

2. Zero budget or space violations in any returned bundle.
3. Runs with one command, no API key required (fallback parser).
4. Evaluators can understand the project in under 60 seconds from the README.

Non-goals (v1)

- Real checkout / cart, live Kohler pricing API, 3D rendering, CAD export.

4. Users

Persona	Need
Homeowner (primary)	"Tell me what fits my room and budget"
Interior designer	Fast, defensible first-pass layout to show a client
Kohler retail / experience-centre staff	A guided-selling tool to use with walk-in customers

5. Scope and functional requirements

ID	Requirement	Priority	Status	Eval criterion
FR-1	Parse NL prompt into $\{\text{length, width, budget, theme, household_size}\}$ using an LLM with strict JSON schema; fall back to regex if no key or on failure	P0	 regex only	Approach & Innovation
FR-2	Structured inputs (sliders / dropdowns) as an alternative to free text	P1		Usability
FR-3	Solver picks one item per category (toilet, shower, vanity, faucet) with total cost $\leq$ budget	P0	 see Known Issues	Technical Execution
FR-4	Solver enforces total footprint $\leq$ 40% of floor area	P0		Technical Execution
FR-5	Solver enforces clearances (toilet side $\geq$ 15 in, front $\geq$ 21 in; shower/vanity front clearance) as placement rules	P0		Technical Execution
FR-6	If no valid bundle exists, return a clear message and the closest alternative (never silently return an invalid one)	P0		Technical Execution
FR-7	Theme matching (Japandi / Minimalist Modern / Classic Luxury) and finish matching across items	P1	 theme only	Approach & Innovation
FR-8	2D floorplan at true scale: room walls, door swing, fixtures at real dimensions, clearance zones, no overlaps	P0	 fixed positions	Approach & Innovation
FR-9	Itemised BOM with SKU, price, dimensions, finish, total	P0		Technical Execution
FR-10	Space-utilisation score	P1		Approach & Innovation
FR-11	Annual water use vs legacy baseline, driven by household size	P1	 formula needs fix	Approach & Innovation
FR-12	Wellness tags per product (e.g. Dekoda-compatible, circadian lighting), shown as labels	P2	 data only	Approach & Innovation
FR-13	Export: JSON config, printable spec sheet	P2		Usability

ID	Requirement	Priority	Status	Eval criterion
FR-14	Multi-page UI: Home, Planner, Sustainability, BOM	P1		Usability

6. Non-functional requirements

Area	Requirement
Performance	End-to-end response < 2 s (local catalog, no DB)
Reliability	<code>pip install -r requirements.txt && python app.py</code> works on a clean machine
Determinism	Same input → same bundle (LLM at temperature 0, solver has no randomness)
Testability	<code>python -m pytest tests/</code> covers solver constraints and parser fallback
Accessibility	Colour contrast ≥ 4.5:1, keyboard-focusable controls
Responsiveness	Usable from 360 px to 1440 px width

7. Architecture



8. Data model (catalog.json)

Each product: `{id, name, category, sku, price_inr, dimensions_in{width,depth,height}, clearance_req_in{side_min,front_min}, gpf | gpm, aesthetic_themes[], finishes[], smart_features[], health_features[], source_url}`.

Requirements on the data:

- Minimum **4 products per category** (16+ total) so the solver has real choices.
- Prices in **INR**, matching kohler.co.in.
- Every SKU verified against kohler.co.in; `source_url` filled in.

9. Information architecture

Page	Route	Purpose
Home	<code>/</code>	Hero, how it works in 3 steps, theme cards, CTA
Planner	<code>/planner</code>	Prompt + sliders, live floorplan, metrics
Sustainability	<code>/sustainability</code>	Water comparison chart, wellness tags
Bill of Materials	<code>/summary</code>	Itemised list, totals, export

10. Design requirements

- Visual language follows kohler.co.in: light, white-dominant, black type and CTAs, generous whitespace, restrained accent colour.
- Floorplan drawn as an architectural drawing: thin black walls, light-grey fixtures, dashed clearance zones.
- Errors and violations shown in plain language, never hidden.
- Design tokens live in `static/css/kohler.css` (single source of truth).

11. Success metrics

Metric	Target	How verified
Budget violations	0 across 50 random test inputs	<code>tests/test_solver.py</code>
Space violations	0 across 50 random test inputs	<code>tests/test_solver.py</code>
Parse success (valid JSON)	100% incl. fallback	<code>tests/test_parser.py</code>
Latency	< 2 s	manual + logged in response
Cold-start setup	≤ 3 commands	README quick start

12. Known issues in current build (to fix before submission)

#	Issue	Impact	Fix
1	Solver picks the priciest affordable item per category, in order; the toilet can eat the budget and later categories fall back to over-budget items	Breaks the "0 budget violations" claim	Search combinations (16 items → tiny), pick best-scoring valid bundle
2	Clearance rules are in the catalog and the deck but not enforced in code	Claim not backed by code	Implement FR-5 in <code>layout.py</code>
3	No LLM call exists; parser is regex only	"AI" claim is weak	Implement FR-1 with fallback
4	Water formula adds faucet GPM into the shower-minutes term	Overstated usage	Separate toilet, shower, faucet terms
5	Catalog has too few items; prices in USD	Little to optimise; mismatch with Kohler India	Expand and convert (Section 8)
6	UI ignores <code>budget_compliant = false</code>	Silent failure	FR-6

13. Milestones

When	Deliverable
19 Sept	Repo cleanup, README, fix issues 1, 2, 6, restyle to Kohler palette
20 Sept	LLM parser + fallback, expanded catalog, video, Prompts PDF, 4-slide deck, email sent
21 Sept	Buffer, rehearse demo, prepare interview answers
22-23 Sept	Interviews, Kohler office, Kharadi
After	Multi-page UI polish, drag-and-drop layout, export

14. Submission deliverables

Deliverable	Location	Status
Working code	GitHub repo	
Prompts documentation	<code>docs/Prompts_Documentation.pdf</code>	
4-slide pitch deck	<code>docs/Presentation_Deck.pdf</code>	
1-3 min demo video	link in README	
Submission email	to Kohler contact, subject <code>AI Lab KOHLER-MITWPU Case Study - (PRN_Name)</code>	

15. Risks

Risk	Mitigation
LLM API fails during live demo	Regex fallback is automatic; demo works offline
Wrong SKU or price spotted by Kohler staff	Verify each item on kohler.co.in, include <code>source_url</code>
UI rebuild breaks working demo	Do it on a branch, merge only when tests pass
Time (deadline 20 Sept)	Priority order in Section 13; multi-page polish is post-deadline